Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Process development for processing high-strength aluminum alloys using tailor welding technology and tailor heat sheet metal forming for innovative lightweight components*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		*Mechanical・Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile / Rail Vehicles	*Vehicle Body and Light-weighting Technologies*	*Plastic Deformation/ Powder*	*Sheet Forming Technologies*
Concept of Proposed Technology	*Increased degree of lightweight construction and improved manufacturability of automotive structural components through the use of new high-strength aluminium alloys.*			
Rationale for Joint R&D Project	*The use of new high-strength natural hard aluminium alloys (HTF: heat treatable free) allows for drastic weight savings in vehicle construction and is therefore particularly interesting for e-mobility. The Korean automotive industry has so far specialized mainly in steel materials processing. Therefore, there is considerable potential for optimization with regard to lightweight aluminum components, particularly in the processing of high-strength aluminum alloys for safety-related car body components. Consistent lightweight construction can improve vehicle weight and thus energy efficiency. Fraunhofer Institutes have world-leading expertise in technology development for welding and forming of high-strength aluminium alloys and transferring this expertise to German*			

	and European car manufacturers.
Key Words	1. *Manufacturing of high-strength aluminium alloys for lightweight construction applications* 2. *Innovative laser welding process for high-strength aluminium alloys* 3. *Innovative forming process with tool-integrated heat treatment prior to forming*.

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (7)
R&D Objectives	*New high-strength aluminum alloys offer the possibility of increasing lightweight construction in automotive structural components. With current processes, these alloys can only be joined and formed to a limited extent due to their increased susceptibility to cracking. Fraunhofer IWS and Fraunhofer IPT are working on a new machining process for this purpose.*			
R&D Target Specifications	*Material-specific manufacturing technologies ready for transfer to Korean automotive production for car bodies and chassis: (1) process-monitored laser welding of aluminium tailor welded blanks and (2) temperature-controlled selective heat treatment for tailor annealed blanks (TAB) for sheet metal forming. Demonstration in a relevant environment and provision of all necessary technology and material data.*			

R&D Plan

Specific Works	Institution	1st Year				2nd Year				3rd Year				Final Year	
		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
1. Development of TWB prototype	Hwashin														
1-1. Integrated structure design															
1-2. Forming analysis, considering Almag material															
1-3. In-line forming characteristic analysis															
1-4. Forming Try-Out															
1-5. Design to optimize energy absorption															
1-6. performance evaluation															
3. Tailoring aluminum sheets	KITECH														
3-1. Tailoring high elongation sheets															
3-2. Tailoring high strength sheets															
3-3. Meterial evaluation															
3-4. Formability evaluation of raw and TWB sheets															
3-5. Formability evaluation of raw and TWB sheets															
4. Process development for TWB / TAB	Fraun-hofer														
4-1. Requirements analysis and process routes															
4-2. Development of TAB technology															
4-3. Development of TWB technology															

4-4. Process design and iterations																
4-5. Design and process optimization for TAB/TWB, interaction of welding and forming process																
4-6.: Evaluation in the progressive die																

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Fraunhofer IWS: Leading expertise in precise laser welding technology for high-strength aluminum alloys. Development of material-adapted laser welding technology for tailor welded blank (TWB) manufacturing.* *Fraunhofer IPT: High level of expertise in thermally assisted sheet metal forming and local soft annealing. Development of Tailor Annealed Blanks (TAB) technology integrated into the forming process through integrated heating in the forming tool (progressive die). As well as the development of local thermally assisted sheet metal processing of automotive components for local softening to increase formability, followed by evaluation in the progressive die on a servo press line (120 strokes/min).*
Expected R&R of Korean SME Partner	Hwashin: (https://www.hwashin.co.kr/eng/main.do): *HWASHIN specializes in manufacturing chassis & body parts.* *Hwashing has good knowlege for Stamping process is one of the most representative processes of Hwashin and has Hot Stamping Press, 2700Ton Transfer, 2000Ton Transfer, 1700Ton Transfer, 1200Ton and several small lines to produce stamping parts.* *And also Hwashin's welding process conducted to assemble Cross Member, Rear CTBA, and Lower Arm with press products and well equipped characteristics of this process for steel products and advanced study for AL stamping and extrusion parts.* KITECH (https://eng.kitech.re.kr/main): Inventor institute of innovative high-formable aluminum alloys with high magnesium contents of 6~10wt% (Almag), which have potential benefits of lower density, higher formability due to stacking fault energy control, higher strength due to work hardening, and generally heat-treatment-free (HTF) nature.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Currently, TWB and TAB processes for aluminum plates have been limited and tried only on heat-treated aluminum sheets with single properties, but the application of TWB and TAB using heat-treatment-free (non-heat-treated or work-hardened) sheets that can achieve various properties, such as steel plates, can provide more innovative design freedom. TWB and TAB are broadly welding based technologies. However, it is challenging to control and optimize melting and solidification to advance the quality of TWB sufficiently. Furthermore, it is necessary to control and optimize local softening to achieve local formability of TAB for subsequent forming processes.The project develops innovative, heat treatment-free aluminum alloys as well as tailor-welded and tailor-annealed blank technologies for the production of high-performance lightweight components for future mobility. Targeted local adaptation of strength and formability reduces weight, costs, and scrap while significantly improving design freedom, crash safety, and energy efficiency. Close cooperation with Fraunhofer Institutes accelerates technology transfer and strengthens the international competitiveness of the Korean mobility and materials industry.*

6. References

(1) Verma R.; Taneja T.; (2024) Aluminum Alloys Consideration in Lightweighting Vehicle Structural Components: Material Selection and Design Considerations Journal of Advanced Research in Automotive Technology & Transportation System.

(2) Chen J.; (2023) Research on Aluminum Alloy Materials and Application Technology for Automotive Lightweighting" by Jianru Chen, Journal of Materials & Chemistry

(2)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization